

Varun Patil

varunpatil@ucla.edu | linkedin.com/in/varun-patil | github.com/pulsejet

EDUCATION

University of California, Los Angeles

Ph.D. Computer Science (Advisor: Lixia Zhang)
M.S. Computer Science

California, USA

Dec. 2020 – Jun. 2025

Sep. 2020 – Dec. 2021

Indian Institute of Technology Bombay

B.Tech. Mechanical Engineering
Institute Technical Roll of Honour

Mumbai, India

Jun. 2016 – Aug. 2020

PROFESSIONAL EXPERIENCE

Meta Platforms, Inc.

Research Scientist – WhatsApp

Menlo Park, California

Jun. 2025 – present

- Building Meta's Private AI inference platform, secured by Trusted Execution Environments (TEEs)
- Designed and implemented the secure and transparent end-to-end data plane used for TEE communication
- Designed and reviewed network communication protocols with strong security requirements

Software Engineering Intern – WhatsApp

Apr. 2019 – Jun. 2020

- Implemented the AMD SEV-SNP Remote Attestation library powering all TEE deployments at Meta
- Developed and deployed integrations for multiple new and existing large-scale services at WhatsApp

Operant Networks

Software Developer Co-Op (C++) – Security

Los Angeles, California

Jun. 2023 – Jun. 2025

- Implemented a cross-platform TPM-backed key storage solution for the Defined-Trust Transport (DeftT)
- Designed a new testing framework, internal tooling based on Conan, and 3× faster cross-platform CI/CD pipelines
- Designed a retrofit field device remote access platform with flexible ACL controls

LabN Consulting, L.L.C.

Software Engineering Intern (C/C++) – LCPS

North Bethesda, Maryland

Jun. – Sep. 2022

- Worked on porting a legacy implementation of the RSVP-TE protocol to the FRRouting software suite
- Ported the legacy code and tests to run on 64-bit hardware and implemented shims for integration with FRR

Computer Centre, IIT Bombay

Full-Stack Developer (TypeScript) – Single Sign On

Mumbai, India

Sep. 2019 – Sep. 2020

- Implemented an OAuth2/OIDC/SAML provider for authorization over LDAP-TOTP-FIDO authentication
- Implemented a very high performance geolocation provider (GeoIPNS) for suspicious activity detection
- Built popular open source plugins for migrating Nextcloud and Roundcube webmail to OpenID Connect
- Integrated authentication with SAML federated identity for Azure Active Directory and Google Workspace

Mercari, Inc.

Backend Software Engineering Intern (Go) – Product Catalog

Tokyo, Japan

Jul. 2019

- Developed a scalable backend for an internal data management tool for an arbitrarily large dataset
- Set up continuous integration and deployment over Kubernetes using CircleCI and Terraform

RESEARCH EXPERIENCE

UCLA

Graduate Student Researcher – Internet Research Laboratory

Los Angeles, California

Apr. 2021 – Jun. 2025

- Designed the State Vector Sync transport protocol for Named Data Networking, a future internet architecture
- Designed NDN Distance Vector (ndn-dv), a highly efficient multi-path routing protocol for named data networks
- Designed the security and transport layer protocols of Ownly, a decentralized NDN workspace application
- Implemented and optimized NDNd, a high performance user-space named packet forwarder in Golang
- Built a friendly network and security simulator-visualizer for running experiments in the browser

IIT Bombay

Undergraduate Student Researcher – Machine Learning Lab, Networked Systems Lab

Mumbai, India

2018 – 2020

- Built data pipelines and improved model architecture for grammatical error correction with deep learning
- Implemented and deployed an automated integration testing framework with Jenkins for a 5G packet core

SELECTED PUBLICATIONS

- [1] **Patil**, Desai, Zhang, 2022. *Kua: A Distributed Object Store over Named Data Networking*. ACM ICN '22
- [2] Moll, **Patil** et al. 2022. *The Evolution of Distributed Dataset Synchronization Solutions in NDN*. ACM ICN '22
- [3] **Patil**, Song, Xiao, Zhang, 2022. *Scaling State Vector Sync*. ACM ICN '22
- [4] Moll, **Patil**, Zhang, Pesavento, 2021. *Resilient Brokerless Publish-Subscribe Over NDN*. IEEE MILCOM '21
- [5] **Patil**, Moll, Zhang, 2021. *Supporting Pub/Sub over NDN Sync*. ACM ICN '21
- [6] Moll, **Patil**, Sabharwal, Zhang, 2021. *A Brief Introduction to State Vector Sync*. TR-NDN-0073

OPEN SOURCE PROJECTS

- Memories** | *PHP, Vue* | 3k+ GitHub stars 2022 – 2024
- Efficient and feature-rich photo management app for Nextcloud, supporting large media libraries
 - First FOSS self-hosted photo manager to have all basic features of commercial alternatives
- NDN-Play** | *Angular, NDNts* 2021 – 2023
- Network simulator and visualizer for running a Named Data Network completely in the browser
 - Features logic for packet routing and forwarding, topology manipulation and live traffic visualization
 - Features a code editor with syntax completion; lets the user run real code on the simulated nodes
 - Supports integration with Mininet, acting as a web interface to an emulated network topology
- Imperials** | *Golang, React* 2021 – 2022
- FOSS web alternative to the Settlers of Catan board game with beautiful graphics
 - Features a rule-based AI supporting player interaction that can play all game modes
- InstiApp** | *Django, Angular, Android, Flutter* 2018 – 2020
- Open source university student activity platform for events, organizations, news, food, maps, etc.
 - Implemented functionality for notifications, a search engine and hierarchical permissions and access control
 - Advanced features include event advertisements to targeted groups, users interests, achievements etc.
 - Led the original team of 10+ developers; the IIT Bombay deployment has 8000+ active users

TECHNICAL SKILLS

Advanced: JavaScript/TypeScript, Python, C/C++, PHP, Golang, Docker
Intermediate: C#, Ruby, Java, C, Lisp, Lua, Kubernetes